

Antonio Ferrão Neto

THE METAPHOR OF THE BUBBLELAND

A PROTOSCIENTIFIC APPROACH

Appendices: Methods and Formalism



Volume III

 **ImpNet**

THE METAPHOR OF THE BUBBLELAND

A Protoscientific Approach

Antonio Ferrão Neto

VOLUME III

Appendices: Methods and Formalism



© 2026 Antonio Ferrão Neto

All rights reserved. No part of this work may be reproduced, stored in a retrieval system or transmitted, in any form or by any means — electronic, mechanical, photocopying, recording or otherwise —, without the prior written permission of the author, save for the statutory exceptions for free use, among them quotation in academic work with due credit, under art. 46 of Brazilian Law 9,610/98.

Exception — openly licensed material. The **figures**, the **code listings** and the **cover images of all three volumes** are not covered by the reservation above: they are released under a **Creative Commons Attribution 4.0 International** license (CC BY 4.0) and may be reproduced, adapted and used freely, including commercially, provided the source is credited — *The Metaphor of the Bubbleland*, by Antonio Ferrão Neto. The figures and listings can be downloaded from the *QR codes* in the captions or directly at
<<https://www.impnet.com.br/bolholandia/en/>>.

Typeset with LaTeX using the ABNT template.

Dados Internacionais de Catalogação na Publicação (CIP)
(Câmara Brasileira do Livro, SP, Brasil)

Ferrão Neto, Antonio

The Metaphor of the Bubbleland : a protoscientific approach - Appendices: Methods and Formalism / Antonio Ferrão Neto ; [tradução do autor]. -- São Bernardo do Campo, SP : ImpNet, 2026. -- (The metaphor of the Bubbleland ; 3)

Título original: A metáfora da Bolholândia.

Bibliografia.

ISBN 978-65-977067-4-7

1. Biologia molecular 2. Computação 3. Equações diferenciais 4. Física 5. Mecânica quântica
I. Título. II. Série.

26-403410.0

CDD-530.12

Índices para catálogo sistemático:

1. Mecânica quântica : Física 530.12

Eliete Marques da Silva - Bibliotecária - CRB-8/9380

About the cover

In the foreground, the cover shows a symbolic representation of a hybrid device: an ancient **Babylonian tablet** (Figure 2) merged with the idea of modern hardware. On it one reads a principle of calculation that *recalls* the **Successive Zooms Method** — one of the pillars of this work and the central theme of this volume (Appendix X). In the background, the trajectories of **subatomic particles** allude to the discussions on particle physics, also gathered here, in the second half of the volume.

This cover image was composed with the aid of artificial intelligence. It builds on the photograph of tablet YBC 7289 by A. Urcia, of the Yale Peabody Museum of Natural History, in the public domain (CC0), obtained via Wikimedia Commons. The final artwork is released under a **Creative Commons Attribution 4.0** license (CC BY 4.0): it may be reproduced, adapted and used freely, including commercially, provided the source is credited — *The Metaphor of the Bubbleland*, by Antonio Ferrão Neto.



Note to the reader

This is the third of the three volumes that make up *The Metaphor of the Bubbleland*. Whereas Volume I gathers the chapters of the work, this volume III, like volume II, is devoted exclusively to the appendices, bringing together the complementary material that supports the argument developed in the chapters.

Here the reader will find, in the very first appendix, a discussion of the Successive Zooms Method, a new numerical/computational approach based on hybrid analog/digital hardware for solving the “stubborn problems” presented in appendix V (Vol. II) — Types of “stubborn problems” in numerical computation. Appendix Z — The urgency of the SZM shows the importance of carrying out research in this area. Appendix AA — Hypotheses for medicine highlights possible applications in Medicine, which makes such research all the more pressing. And appendix AC — Suggestions for using the SZM in Molecular Dynamics problems shows how to apply the Successive Zooms Method within a Molecular Dynamics program (GROMACS), with an extension for the study of another hypothesis addressed in the book, namely the “Tunneling of Biomolecules” (Chapter 8, Vol. I — Tunneling of biomolecules). The volume also includes several appendices on particle Physics.

Throughout Volume I and Volume II, indications of the type “see Appendix . . .” refer directly to the sections gathered here. Such indications may also occur among the appendices themselves. Each appendix is identified by a letter and a title of its

own, and can be located through the table of contents in the opening pages of the respective volumes.

Short Table of Contents

About the cover	1
Note to the reader	3
Short Table of Contents	5
Table of Contents	7
X Brief introduction to the Successive Zooms Method (SZM)	13
Y Analog and hybrid computation for PDEs	107
Z The urgency of the SZM	135
AA Hypotheses for medicine	207
AB Taxonomy of black swans	229
AC Suggestions for using the SZM in Molecular Dynamics problems	233
AD The evolution of artificial intelligence	301
AE Animation, Modeling, and Molecular Dynamics	317
AF Internal and formal structures of particle physics	329
AG Listing of some programs	341
AH Mapping of hypotheses	391
AI Violation of time symmetry	409
AJ Heuristic derivation of the Schrödinger equation	413
AK Difficulties in computing the equilibrium of bubbles	421
AL Techniques for the structural determination of biomolecules	427

AM Is it possible to plot an electron from the Dirac equation?	437
AN The dominance of Mathematics in the Standard Model	445
Bibliography	449
Index	489

Table of Contents

About the cover	1
Note to the reader	3
Short Table of Contents	5
Table of Contents	7
X Brief introduction to the Successive Zooms Method (SZM)	13
X.1 Description of the method	17
X.2 Definition of an SZM device (SZMD)	26
X.3 General idea of the program <code>szm_device.py</code>	34
X.4 Feasibility simulation <code>szm_feasibility.py</code>	37
X.5 SZM speed versus digital precision	39
X.6 Richardson extrapolation/analysis applied to the SZM	52
X.7 Implicit methods, hybrid methods and the verification of the SZM results	56
X.8 SZM and other numerical methods on digital computers	59
X.9 The SZM in quantum computing	68
X.10 The extended analog computer (EAC)	70
X.11 Partial differential equations (PDEs) and the SZM	72
X.12 Some limitations of the general-purpose analog computer	73
X.13 Solving PDEs using FPGAs	84
X.14 From differential equations to successive zoom architectures	86
X.15 Continuous blocks versus discrete meshes in solving PDEs	88
X.16 Analog computation for accelerators	89
X.17 Huang's (2017) hybrid analog-digital solution of nonlinear PDEs	92
X.18 Stochastic (Monte Carlo) method for PDEs on analog computers	95
X.19 Digital differential analyzers (DDA)	96
X.20 Analysis of the MacLennan (2007) article	98
Y Analog and hybrid computation for PDEs	107
Y.1 Frequency-based analog computation	108
Y.2 Introduction and conceptual foundations	115

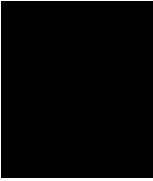
Y.3	Importance of the hybrid architecture in the SZM	123
Y.4	Optical and photonic computation	124
Y.5	SZM, nanotechnology, and problems of astronomical dimension	127
Y.6	Why the SZM hardware details fall outside this book	134
Z	The urgency of the SZM	135
Z.1	Why the urgency?	136
Z.2	The urgency of knowledge about the SZM	136
Z.3	Analytic but numerically intractable	146
Z.4	Some examples of problems related to life	149
Z.5	Multiscale computational modeling with the SZM	192
Z.6	Conclusion	204
AA	Hypotheses for medicine	207
AA.1	The dreams of modern medicine	208
AA.2	The undreamed dreams of modern medicine	210
AA.3	Therapeutic applications of generative genomic models	211
AA.4	Cellular communication by electromagnetic radiation	213
AB	Taxonomy of black swans	229
AB.1	From the problem of induction to the black swan	230
AB.2	Four stances toward the black swan	230
AB.3	Operational principle	232
AC	Suggestions for using the SZM in Molecular Dynamics problems	233
AC.1	Introduction	236
AC.2	Integration architecture	239
AC.3	Interface specification	240
AC.4	Stub implementation for prototyping	254
AC.5	Integration with the GROMACS MD loop	276
AC.6	Considerations on external hardware	282
AC.7	AI in SZM-GROMACS integration	284
AC.8	Compilation and installation	286
AC.9	Complete usage example	287
AC.10	Extension to 4D+T simulations	288
AC.11	Limitations and future work	295
AC.12	Conclusion	297
AC.13	The evolution of abstraction	298
AD	The evolution of artificial intelligence	301
AD.1	Timeline of artificial intelligence	302

AD.2 The two paradigms: symbolic versus connectionist	303
AD.3 The discredit of the pioneers and the AI winters	304
AD.4 The theoretical turning point: backpropagation	305
AD.5 Why modern AI could not have emerged earlier	306
AD.6 The era of transformers and generative models	307
AD.7 Conceptual synthesis	308
AD.8 Methodological lesson learned	308
AD.9 LLM-based AIs and the discovery of nomies	310
AE Animation, Modeling, and Molecular Dynamics	317
AE.1 Molecular Animation	318
AE.2 Molecular Modeling	320
AE.3 Molecular Dynamics	321
AE.4 Molecular Modeling versus Molecular Dynamics	323
AE.5 The case of <i>docking</i> and its variants	324
AE.6 The molecular Monte Carlo method	325
AE.7 Comparative table	327
AE.8 Conclusion: three activities, three epistemologies	327
AF Internal and formal structures of particle physics	329
AF.1 Symmetry groups and state spaces	330
AF.2 The $U(1)$ group and the structure of the membrane	331
AF.3 The $SU(2)$ group and spinors	332
AF.4 The $SU(3)$ group and confining structures	333
AF.5 Generalizations and the role of $SU(N)$	334
AF.6 Limits of the metaphor	334
AF.7 Structure of the QED Lagrangian	335
AF.8 Nonlinearity of electromagnetism	336
AF.9 Scalars, vectors, tensors, and spinors	337
AF.10 Synthesis: from the conceptual to the formal	339
AG Listing of some programs	341
AH Mapping of hypotheses	391
AH.1 Legend and classification criteria	392
AH.2 Taxonomy of the central hypotheses	393
AH.3 Master table of the hypotheses	393
AH.4 Analysis by epistemic strength	393
AH.5 Network of dependencies among the hypotheses	401
AH.6 Clusters of mutually reinforcing hypotheses	401

AH.7 Hypotheses with transformative impact	406
AH.8 Hypotheses at greatest risk of refutation	407
AH.9 Synthesis	407
AI Violation of time symmetry	409
AI.1 The fundamental symmetries C, P and T	410
AI.2 The breaking of the symmetries: from Wu to BaBar	410
AI.3 Limits and synthesis	411
AJ Heuristic derivation of the Schrödinger equation	413
AJ.1 The d'Alembert equation	414
AJ.2 From wave equation to matter wave	415
AJ.3 Classical energy as a starting point	416
AJ.4 General form in three dimensions	417
AJ.5 Conceptual rupture	417
AJ.6 From the Schrödinger equation to quantum electrodynamics	418
AJ.7 From QED to the Standard Model	418
AK Difficulties in computing the equilibrium of bubbles	421
AK.1 Soap bubbles as analog computers	422
AK.2 The computational infeasibility of digital simulation	422
AK.3 From bubble to particle: a leap in complexity	423
AK.4 A force that does exist between bubbles: the Cheerios effect	424
AL Techniques for the structural determination of biomolecules	427
AL.1 X-ray crystallography	428
AL.2 Nuclear magnetic resonance (NMR)	428
AL.3 Cryo-electron microscopy (cryo-EM)	429
AL.4 Structural mass spectrometry	429
AL.5 Atomic force microscopy (AFM)	430
AL.6 Vibrational spectroscopies (Raman and infrared)	430
AL.7 Advanced fluorescence microscopy	430
AL.8 Small-angle scattering (SAXS / SANS)	431
AL.9 Electron crystallography (MicroED)	431
AL.10 Computational structural modeling	432
AL.11 Comparative table	432
AM Is it possible to plot an electron from the Dirac equation?	437
AM.1 Why the 720-degree rotation?	439
AM.2 What the “electron drawings” really show	440
AM.3 A concrete case: the Dirac hydrogen atom	441

AM.4 The deeper obstacle: the electron is not a wave function	441
AM.5 Conclusion: the map is not the territory	443
AN The dominance of Mathematics in the Standard Model	445
AN.1 When symmetry defines ontology	446
AN.2 Other areas: from observation to equation	446
AN.3 Philosophical consequences	447
 Bibliography	 449
Index	489

APPENDIX AH



Mapping of hypotheses

I would rather have questions that cannot be answered than answers that cannot be questioned.

— AUTHOR UNKNOWN

THIS appendix presents a systematic survey of the hypotheses formulated throughout the book, classified by epistemic strength, nature, location, and derivation relationships.

The reader will notice that most of the hypotheses cataloged here receive a *weak* classification when assessed in isolation. This is intentional: each hypothesis, taken independently, lacks direct empirical support or a detailed mechanism.

It should be acknowledged that such classifications are *inherently subjective*. There is no universally accepted criterion for quantifying the “strength” of a speculative

hypothesis, and another evaluator — with different philosophical assumptions or standards of evidence — could legitimately reach different conclusions. The mapping presented here reflects the author’s judgment, not an objective verdict.

AH.1 Legend and classification criteria

Table 32 – Criteria for classifying the epistemic strength of the hypotheses.

Symbol	Level	Description
S	Strong	Grounded in established physics, with serious historical precedents or broad experimental support
M	Moderate	Plausible extension of current knowledge, testable in principle, with partially specified mechanisms
W	Weak	Speculative or protoscientific, with no detailed mechanism or immediate test available

Several of these weak hypotheses are *collaborative* — they reinforce one another, share assumptions, and, when combined, form an argumentative structure whose internal coherence grants the set a greater epistemic strength than the sum of its parts. A detailed discussion of the mechanisms by which individually weak hypotheses can strengthen one another when articulated into coherent networks — including consilience, structural coherence, inference to the best explanation, Bayesian analogy, mutual constraint, heuristic fertility, the limits of reinforcement, and the problem of empirical underdetermination — is developed in Section B.3 (Vol. II), in Appendix B (p. 73) — Types of hypotheses. The *criteria of legitimate strengthening* **Cf1–Cf8** established there — complementary to the criteria of individual admissibility **Cr1–Cr5** (Section B.2, Vol. II, p. 78) — apply to the analysis of the hypothesis clusters developed in the subsequent sections of this appendix.

Another relevant point about the hypotheses classified as *weak*, in the context of this book, is that several of them seek to explain currently inexplicable facts — that is, they aim to fill gaps in established Science. The absence of alternative hypotheses for these same phenomena is not evidence in their favor — that would be arguing from ignorance — but it grants them a place: lacking competitors, even a weak hypothesis *provisionally* occupies an explanatory space that would otherwise remain empty, and only for as long as it remains refutable in principle (criterion Cr5, Section B.2.1,

Vol. II).

AH.2 Taxonomy of the central hypotheses

The hypotheses can be grouped into six broad thematic classes:

1. **Dimensional and geometric** — Existence of additional dimensions and parallel realities.
2. **Ontological** — Internal structure of particles, nonlinearity of fields, nature of the fundamental Lagrangian.
3. **Biomolecular** — Tunneling and coupling with parallel realities.
4. **Temporal** — Temporal disconnection between realities; time travel; emergent nature of time.
5. **Mind and consciousness** — Noncomputable nature; interface with external realities.
6. **Methodological** — Discovery of nomies; SZM as a multiscale bridge.

AH.3 Master table of the hypotheses

Table 34 presents the complete survey. The columns indicate:

- **ID**: unique identifier (H + sequential number);
- **Hypothesis**: synthetic description;
- **Class**: thematic category (1–6, as above);
- **Strength**: level of support (S/M/W);
- **Origin**: chapter/appendix where it is introduced;
- **Testability**: experimental paths where applicable.

Note on the “Class” column (“Cl.”): Most of the hypotheses belong to a single thematic class. In specific cases of genuine membership in two classes — for example, hypotheses about cognitive limits whose origin is nomological-dimensional (H02, H04), or about mental processing whose location involves parallel realities (H23, H24) — the notation **X/Y** is used, indicating the primary class followed by the secondary class.

AH.4 Analysis by epistemic strength

The distribution of the hypotheses by epistemic strength (according to the subjective classification discussed in the introduction) reveals the overall profile of the work:

Table 33 – Distribution of the hypotheses by epistemic strength.

Level	Count	Percentage
Strong (S)	2	4.35%
Moderate (M)	10	21.74%
Weak (W)	34	73.91%
Total	46	100%

The predominance of weak hypotheses (74%) is consistent with the protoscientific character of the work, as stated from the Preface onward. The set of moderate hypotheses (22%) is concentrated in methodological propositions (SINDy, SZM) and in vocabulary reorganizations that do not alter experimental predictions (formalization of the $\otimes$ operator, decomposition into subnomies). The two hypotheses classified as strong are H05 (“the Standard Model is an effective theory”), which corresponds to an already established consensus in contemporary theoretical physics, and H25 (“consciousness is not reducible to algorithmic computation”), which has solid grounding in Gödel’s incompleteness theorems and in the limits of the Turing Machine, as developed by Penrose (PENROSE, 1989; PENROSE, 1994). Two clarifications about what the table contains. First, it lists the hypotheses this book *adopts*, even when the idea originated with others; proposals by others that the book merely reports — Fröhlich’s coherent modes, Luca Turin’s vibrational theory of olfaction — are not included. Second, the name in parentheses credits the *statement as the row phrases it*, and nothing more: Penrose holds that the mind is not reducible to algorithmic computation, not that it operates outside the brain — a reading he rejects, as Section 5.2 (Vol. I) records; Barbour and Rovelli hold that time is not a fundamental entity, not that it is detuned across realities. What the clusters of this appendix do with those statements is this book’s responsibility, not that of the authors credited.

Table 34 – Master table of the hypotheses presented throughout the book.

ID	Hypothesis	Cl.	Str.	Origin	Testability
H01	Realities can coexist in parallel planes separated by additional dimensions	1	W	Ch. 1 (Vol. I), Ch. 3 (p. 55)	Indirect detection via residual gravitational coupling
H02	Human perception is structurally limited to its own nomies	1/5	M	Ch. 1 (Vol. I), Section 1.3.3.6 (p. 41)	Cognitive studies on semantic categories
H03	A 5D manifold foliated by parallel 4D realities exists	1	W	App. E (Vol. II)	Gravitational anomalies in high-precision experiments
H04	Parallel realities may be non-geometric (informational, conscious)	1/5	W	App. F (Vol. II)	Conceptual test; no direct experimental protocol
H05	The Standard Model describes an effective theory, not an ultimate description	2	S	Ch. 11 (Vol. I), Section 11.6 (p. 225)	Theoretically consolidated; search for physics beyond the SM
H06	Fundamental particles are nomions — localized structures of nonlinear fields	2	M	Ch. 12 (Vol. I), Section 12.1 (p. 237)	High-energy scattering; contributions to $g - 2$
H07	The electric field is nonlinear in the high-intensity regime near the electron	2	W	Section 12.5 (Vol. I)	Precision tests of Coulomb's law at the microscopic scale
H08	The strong and weak interactions may emerge as effective nonlinear regimes	2	W	Section 12.2 (Vol. I), App. S (Vol. II)	Discovery of a deeper nomy via SINDy + SZM
H09	The quantization of light emerges from nonlinear saturation of the field (alternative to Planck)	2	W	App. M (Vol. II)	High-precision spectroscopy experiments
H10	The equality $ q_e = q_p $ reflects a deep nomological coordination	2	M	App. O (Vol. II)	Tests of matter neutrality at precision $\lesssim 10^{-21}$

Continued on next page...

Table 34 (continued)

ID	Hypothesis	Cl.	Str.	Origin	Testability
H11	The tangible nomy decomposes into subnomies coupled by the $\otimes$ operator	2	M	Ch. 14 (Vol. I), Def. 45 (p. 301)	Vocabulary reorganization; indirect validation by consistency
H12	The electromagnetic and gravitational subnomies are ontologically primitive	2	W	Section 12.2 (Vol. I)	Derivation of $SU(3)_C$ and $SU(2)_L$ as effective regimes
H13	The 1s orbital is a distribution of epistemic ignorance, not real motion	2	W	Subsection 12.12.1 (Vol. I)	Ontological reinterpretation; does not alter QM predictions
H14	Biomolecules can access 4D degrees of freedom under specific conditions	3	W	Ch. 8 (Vol. I), Subsection 8.1.1 (p. 144)	MINFLUX: spatial jumps, anomalous <i>blinking</i>
H15	Biomolecules can transiently shift to a parallel reality of complementary nomy	3	W	Subsection 8.2.2 (Vol. I), Box 26 (p. 149)	Temporary disappearances in single-particle tracking
H16	DNA transiently accesses 4D to resolve topological knots	3	W	App. H (Vol. II)	Anomalous efficiency of topoisomerases II
H17	Microtubules participate in the physical substrate of consciousness	5	W	Subsection 8.3.4 (Vol. I)	Correlation between microtubule stabilization and the anesthetic threshold
H18	There is a continuous coupling between our reality and a complementary one, perceptible only at the biomolecular scale	3	W	Subsection 8.4.2 (Vol. I)	MINFLUX signatures dependent on molecular identity
H19	Time in parallel realities may be detuned from ours	4	W	Subsection 8.4.3 (Vol. I), Ch. 4 (p. 75)	Radioactive isotopes in biomolecules (speculative proposal)
H20	Radioactivity may serve as a probe to detect temporal detuning	4	W	Subsection 8.4.4 (Vol. I)	Apparent half-life of $^{13}\text{N}/^{11}\text{C}$ in biomolecules

Continued on next page...

Table 34 (continued)

ID	Hypothesis	Cl.	Str.	Origin	Testability
H21	Time travel to the past creates new timelines (no grandfather paradox)	4	W	Ch. 6 (Vol. I), Section 7.1 (p. 126)	No direct experimental test; logical coherence
H22	Non-intrusive probes could passively monitor the temporal tree	4	W	Ch. 6 (Vol. I)	No available technology; speculative hypothesis
H23	The human mind operates partly in a domain distinct from the biological brain	5/1	W	Ch. 5 (Vol. I), Section 5.2 (p. 85)	Anesthesia + microtubules; box jellyfish without a brain
H24	Box jellyfish perform visual processing in a domain external to the nerve ganglia	5/1	W	Ch. 5 (Vol. I)	Complex visual behavior with no visible neural substrate
H25	Consciousness is not reducible to algorithmic computation (Penrose)	5	S	Ch. 5 (Vol. I), Ch. 10 (p. 203)	Gödel's theorems; limits of Turing
H26	<i>Qualia</i> constitute an irreducible form of non-propositional knowledge	5	M	Ch. 10 (Vol. I)	Mary's Room argument; philosophical debates
H27	Nomies of arbitrary realities can be discovered from observational data	6	M	Ch. 9 (Vol. I), Section 9.5 (p. 189)	Application of SINDy to experimental data
H28	SINDy can be extended to algebraic structures (Group-SINDy)	6	M	Subsection 9.5.1 (Vol. I)	Computational implementation; validation on known systems
H29	Quantum computing can accelerate the discovery of nomies (Quantum SINDy)	6	W	Section 9.6 (Vol. I)	Depends on quantum hardware not yet available
H30	SZM enables arbitrary precision from low-precision analog devices	6	M	App. X	Construction of an SZMD prototype; validation on benchmarks

Continued on next page...

Table 34 (continued)

ID	Hypothesis	Cl.	Str.	Origin	Testability
H31	SZM can be integrated into Molecular Dynamics packages (GROMACS)	6	M	App. AC	Comparative simulations; precision gain in stiff systems
H32	Self-replicating SZM devices at the molecular scale are viable in principle	6	W	Section Y.5.3	Advanced molecular nanotechnology; long horizon
H33	Quantum entanglement may reflect a class variable in a parallel reality	1	W	Subsection 17.4.7 (Vol. I), Box 51 (p. 342)	No direct test; offers a compatible alternative language
H34	Transverse causality may allow interventions without a path in the observable Plane	4	W	Ch. 17 (Vol. I), Section 17.6 (p. 349)	Metaphysical hypothesis; no experimental protocol
H35	“Hyperevolution”: abrupt evolutionary jumps may involve inter-reality coupling	3	W	App. C (Vol. II)	Fossil record of the Cambrian explosion; inconclusive evidence
H36	Dark matter may belong to inaccessible subnomies or parallel realities	1	W	Ch. 12 (Vol. I), Section 12.16 (p. 284)	Astrophysical observations; no direct detection so far
H37	Dark energy may have a nomological origin in a parallel reality	1	W	Ch. 11 (Vol. I)	Observational cosmology; speculative hypothesis
H38	Primordial biomolecules tunneled into parallel realities, resolving the self-replicating RNA paradox	3	W	App. C (Vol. II), Section C.5 (p. 118)	Origin of life; no direct experimental evidence
H39	There exist parallel realities whose nomies define the behavior of the biomolecules of our reality	3	W	App. C (Vol. II)	Indirect inference via correlations between anomalous biomolecular events
H40	The fundamental Lagrangian may admit no closed analytic form, existing only as an algorithmic procedure	2	W	Ch. 12 (Vol. I), Box 36 (p. 253)	Numerical reconstruction via SZM + SINDy

Continued on next page...

Table 34 (continued)

ID	Hypothesis	Cl.	Str.	Origin	Testability
H41	The 19 free parameters of the SM are determined by the internal geometry of the solitons	2	W	Ch. 12 (Vol. I), Section 12.7 (p. 252)	Direct calculation of solitonic energies via SZM
H42	The 3 generations of fermions correspond to distinct radial excitations or topological sectors	2	W	Ch. 12 (Vol. I), Section 12.7 (p. 252)	Excitation spectrum in extended solitonic models
H43	Inside the solitons, the full gauge symmetry is restored	2	W	Ch. 12 (Vol. I), Subsection 14.5.1 (p. 312)	Analogous to 't Hooft–Polyakov; hard to test directly
H44	The Pauli Exclusion Principle emerges from “onion-layer”-type geometric nonlinearities	2	W	Ch. 12 (Vol. I), Section 12.13 (p. 274)	Numerical reproduction of the periodic-table structure via SZM
H45	Parallel 3D realities coexist along a fourth spatial dimension v	1	W	Ch. 3 (Vol. I)	Detection via biomolecular coupling (H14, H15)
H46	Time is an emergent illusion (Barbour) or a local thermodynamic relation (Rovelli), not a fundamental entity	4	M	Ch. 4 (Vol. I)	Theoretical debate in quantum cosmology; no direct test

Note. For the explicit application of the admissibility criteria **Cr1–Cr5** to each hypothesis in this table, see Subsection B.2.4 (Vol. II) of Appendix B (p. 73).

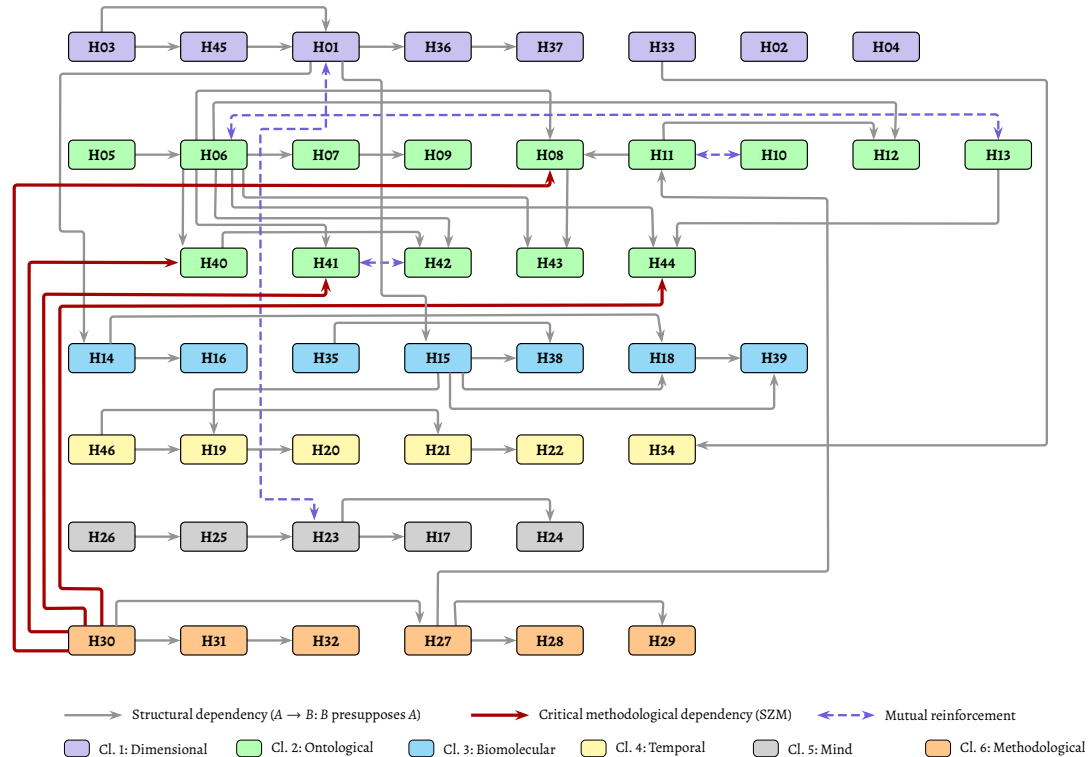


Figure 18 – Network of dependencies among the hypotheses of the book, organized in rows by *primary* class (Table 34); the legend at the foot identifies the class colors and the three stroke types. Hypotheses that belong to two classes (H02, H04, H23, H24; X/Y notation in Table 34) take the color of the first.

